



PROJECT DOCUMENTATION

GridVeil

Privacy-Preserving Electricity Load Forecasting and Anomaly Detection for Libya's Power Grid

A Hybrid CNN-LSTM-Attention Approach to Privacy-Safe Electricity Demand Forecasting Using a Public Regional Proxy Dataset

Prepared by: Team Paradox

Team Members

[Mimuona Jamal Ettaib]

[Fatima Abdulhameed Alshibany]

[Mariam Mahmoud Alawami]

[Nusayba Abdelmutalib Alqaziri]

[25/ September/2026]

TABLE OF CONTENTS

LIST OF TABLES	4
LIST OF FIGURES.....	4
LIST OF ABBREVIATIONS.....	5
ABSTRACT.....	6
 CHAPTER I – INTRODUCTION	 7
1.1 Introduction	7
1.2 Problem Statement	7
1.3 Motivation.....	7
1.3.1 Aim	7
1.3.2 Objectives	8
1.4 Research Questions.....	8
1.5 Scope and Limitations	8
1.5.1 Scope.....	8
1.5.2 Limitations	8
1.6 Methodology.....	9
1.6.1 Research Design	9
1.6.2 Data Description	9
1.6.3 Proposed System Architecture	9
1.6.4 Implementation Details	10
1.6.5 Evaluation Methodology.....	10
 CHAPTER II – INITIATION	 11
2.1 Feasibility Study	11
2.2 Requirements Gathering	11
 CHAPTER III – ANALYSIS	 12
3.1 Software Requirement Specification	12
3.2 Functional Requirements	12
3.3 Nonfunctional Requirements	12
3.4 Introduction to Analysis	13
3.5 System Scenario	13
3.6 Use Case Diagram	13
3.7 Activity Diagram	13
 CHAPTER IV – DESIGN	 14
4.1 Data Model.....	14
4.2 Sequence Diagram	14
4.3 Component Model	14
4.4 Deployment Diagram	14
 CHAPTER V – DEVELOPMENT AND TESTING	 16

5.1 Data Sources and Preparation	16
5.2 System Development	16
5.3 Testing Plan	17
5.4 Testing Results	17
CHAPTER VI – CONCLUSION.....	19
6.1 Summary	19
6.2 Conclusion	19
6.3 Suggestions	19

LIST OF TABLES

Table No.	Title	Section
1.1	Libya installed generation capacity by technology (2018)	1.2
1.2	GridVeil forecasting model configuration	1.6.3
5.1	Final model performance vs. Random Forest baseline	5.4
5.2	Anomaly detection summary	5.4
5.3	Differential privacy experiment results	5.4
5.4	Testing results summary (all stages)	5.4

LIST OF FIGURES

Figure No.	Title	Section
1.1	GridVeil end-to-end data and modelling pipeline	1.6.3
5.1	Training and validation loss curve, Hybrid CNN-LSTM-Attention model	5.4

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory (recurrent neural network)
DP	Differential Privacy
GECOL	General Electricity Company of Libya
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
RMSE	Root Mean Squared Error
R ²	Coefficient of Determination
MW	Megawatt
PII	Personally Identifiable Information
CSV	Comma-Separated Values
GPU	Graphics Processing Unit
UI	User Interface
SRS	Software Requirement Specification

ABSTRACT

Libya's national electricity grid faces a severe and worsening access and reliability crisis, driven by an ageing, fossil-fuel-dependent generation fleet, chronic underinvestment, rapid demand growth and recurring conflict-related disruption. Analysing and forecasting this crisis is made significantly harder by the complete absence of publicly available, granular (sub-hourly) consumption data for the Libyan grid, leaving only sensitive or entirely inaccessible operator records as an alternative. GridVeil is a privacy-preserving analytics and forecasting prototype, developed by Team Paradox, that addresses this gap without compromising data privacy or making unfounded claims about Libya's literal grid behaviour.

The prototype combines two classes of public data: real, national-level Libyan electricity statistics – installed capacity by generation technology, utility structure, distribution losses and renewable-energy targets used to document and visualise the true scale of the crisis; and a publicly available, granular electricity load dataset from Tetouan, Morocco, a North African region with a broadly comparable climate and demand profile, used as an explicitly documented methodological proxy to design, train and validate a forecasting and anomaly detection pipeline. A Hybrid CNN-LSTM-Attention neural network was designed and trained on 10-minute-resolution Tetouan load data using a strict, leakage-free 70/15/15 chronological train/validation/test split and a 4-hour (24-step) lookback window, achieving a test Mean Absolute Percentage Error (MAPE) of 2.69%, a substantial improvement over a Random Forest baseline (5.77% MAPE).

An anomaly detection layer built on a dynamically computed, data-driven threshold (the 95th percentile of the model's training error) flagged 7,834 anomalous observations in the test set, exposing unusual consumption behaviour without relying on fixed, arbitrary cutoffs. A companion Differential Privacy experiment quantified the accuracy cost of protecting individual-level consumption data: introducing Laplace noise at $\epsilon = 1.0$ to a synthetic Libyan pattern load series raised forecasting error from 5.57% to 7.24% MAPE – an approximately 30% relative increase – giving an initial, concrete privacy-utility trade-off measurement. All results are presented through a self-contained, browser based dashboard that visualises Libya's real electricity-system context, the forecasting model's performance against its baseline, detected anomalies, and the outcome of the privacy experiment.

Every dataset used in the project is explicitly classified, sourced and documented, and no personally identifiable or sensitive consumer-level information is used at any stage. The project's central contribution is a reproducible, ethically grounded blueprint: a forecasting and anomaly-detection methodology validated on public proxy data today, ready to be applied directly to Libya's own grid the moment equivalent local data becomes publicly available.

CHAPTER I – INTRODUCTION

1.1 Introduction

Reliable access to electricity underpins economic activity, public services, healthcare and everyday life. In many countries, ageing infrastructure, chronic underinvestment and, in some cases, armed conflict have degraded grid reliability faster than utilities can reinforce or modernise it. Libya is a striking example of this pattern: despite record levels of installed generation capacity, the national grid continues to suffer from frequent, severe outages and a persistent gap between supply and demand. A key barrier to addressing this crisis analytically is data: unlike many national grids, Libya has no publicly available, granular (hourly or sub-hourly) electricity consumption dataset, which makes conventional data-driven forecasting and monitoring approaches impossible to apply directly and transparently.

GridVeil was developed to demonstrate that this data gap does not have to be a dead end. By combining real, public national statistics about Libya's power system with a carefully chosen public proxy dataset for the forecasting methodology itself, it is possible to build and validate a modern, privacy-safe forecasting and anomaly-detection pipeline today, ready to be pointed at Libya's own grid the moment equivalent local data becomes available.

1.2 Problem Statement

Libya's electricity sector is characterised by an ageing, fossil-fuel-dominated generation fleet, chronic underinvestment, rapid population and urban demand growth, and recurring outages linked to conflict and operational instability. The General Electricity Company of Libya (GECOL), the country's sole, fully state-owned utility for generation, transmission, distribution and sales, reported installed generation capacity reaching a record 8,200 MW in 2023 against peak demand of only 5,000–6,000 MW – yet outages remain frequent, pointing to deeper structural and distribution-level problems rather than a simple shortage of capacity. Distribution efficiency is estimated at only around 71%, meaning roughly 29% of distributed electricity is lost before reaching consumers. Compounding this, no public, granular consumption dataset exists for Libya's grid, so any forecasting or anomaly-detection capability that could help plan around these constraints cannot be developed or validated directly against real Libyan operational data without relying on sensitive or inaccessible utility records.

1.3 Motivation

Modern grid operators increasingly rely on data-driven load forecasting and anomaly detection to plan generation dispatch, reduce losses and respond quickly to abnormal conditions. These techniques depend on a foundation of granular, historical consumption data, a foundation Libya does not publicly have. Building a forecasting system directly on sensitive operator data would raise serious privacy and access concerns even if such data could be obtained. This motivates a two-track approach: use what public information does exist about Libya's real system to document the scale of the crisis, and use a comparable, public proxy dataset to build, validate, and stress-test the actual forecasting methodology under realistic conditions, including quantifying what it would cost, in accuracy, to protect that data with formal privacy guarantees.

1.3.1 Aim

To design and prototype a privacy-preserving electricity load forecasting and anomaly-detection system that documents Libya's real electricity crisis using public statistics, and demonstrates a forecasting

methodology on a public, comparable regional proxy dataset, ready for direct application to Libya's own grid once equivalent local data becomes available.

1.3.2 Objectives

- To document and visualise the real scale of Libya's electricity generation, distribution and reliability crisis using vetted public national statistics.
- To identify and justify a suitable public proxy dataset for building a load-forecasting methodology in the absence of granular Libyan data.
- To design and train a hybrid deep-learning model (CNN-LSTM-Attention) for short-horizon electricity load forecasting.
- To implement a data-driven anomaly-detection layer capable of flagging unusual consumption behaviour without relying on fixed, arbitrary thresholds.
- To quantify the accuracy cost of applying formal differential-privacy protection to consumption data through a controlled experiment.
- To present all findings through a self-contained, interactive dashboard that communicates the crisis context, model performance and privacy trade-offs clearly.

1.4 Research Questions

- How can Libya's electricity crisis be documented and communicated using only public, non-sensitive national statistics?
- Can a load-forecasting methodology be validly developed and evaluated on a public proxy dataset from a climatically and demographically comparable region, in the absence of Libya's own granular data?
- How does a hybrid CNN-LSTM-Attention architecture compare to a classical machine-learning baseline (Random Forest) for short-horizon electricity load forecasting?
- Can anomalous electricity consumption be reliably flagged using a threshold derived from the model's own training error distribution, rather than a fixed percentage cutoff?
- What is the measurable accuracy cost of applying differential-privacy noise to consumption data before training a forecasting model?

1.5 Scope and Limitations

1.5.1 Scope

The project covers: the collection and presentation of public Libyan electricity-system statistics; the selection and justification of a public proxy dataset (Tetouan, Morocco) for forecasting methodology development; feature engineering, model design, training and evaluation of a Hybrid CNN-LSTM-Attention forecaster against a Random Forest baseline; a data-driven anomaly-detection layer; a differential-privacy experiment on a synthetic Libyan-pattern dataset; and a browser-based dashboard integrating all of the above. The project does not cover live integration with GECOL's operational systems, real-time data ingestion from an actual grid, or long-horizon (multi-day) forecasting.

1.5.2 Limitations

The forecasting model is trained and evaluated entirely on Tetouan proxy data rather than real Libyan consumption data, so its reported accuracy describes performance on the proxy task, not a guaranteed performance level on Libya's own grid; this substitution is documented explicitly rather than implied. The differential-privacy experiment uses a synthetic, documented Libyan-pattern series rather than real household-level data, since no such data is available or would be ethical to use. The differential-privacy

result reported ($\epsilon = 1.0$) reflects a single privacy budget; a full privacy-utility curve across a range of ϵ values was not exhaustively mapped in this iteration of the project. Testing of the pipeline was carried out through manual functional, integration and end-to-end checks rather than a large-scale automated test suite.

1.6 Methodology

1.6.1 Research Design

This project follows an applied, system-development research design centred on building and evaluating a working prototype rather than an experimental study limited to model benchmarking alone. The work proceeds in two parallel tracks that converge in a single dashboard: a data-documentation track, which curates and visualises real public statistics about Libya's electricity system, and a modelling track, which designs, trains and evaluates a forecasting and anomaly-detection pipeline on a public proxy dataset, then measures the additional cost of formal privacy protection on that pipeline.

1.6.2 Data Description

Four data sources were used, each with a distinct role:

- Libya public national statistics – installed generation capacity by technology and commissioning year, utility structure, distribution losses, and renewable-energy policy targets, used purely for real-world context and visualisation.
- Tetouan (Morocco) load dataset – the proxy time-series dataset used to build and validate the forecasting methodology: 52,417 rows of 10-minute-interval measurements covering January 2017 onward, with temperature, humidity, wind speed, solar-flux features and power consumption for three distribution zones (PowerConsumption_Zone1/2/3).
- Synthetic Libya-pattern electricity data – an hourly, documented synthetic load series with a plausible seasonal and diurnal shape, used exclusively to scaffold and evaluate the differential-privacy experiment without touching any real consumption record.
- Derived engineered features – calendar and behavioural features (hour, minute, day of week, day of year, week of year, month, quarter, season, weekend/working-day flags, business-hour and peak-hour flags, month/quarter start-end flags, minute-of-day and minute-of-week) computed directly from each record's timestamp.

No dataset used in the project contains personally identifiable or individually attributable consumption information; the Tetouan and synthetic datasets are aggregate, zone- or system-level series rather than household-level records.

1.6.3 Proposed System Architecture

GridVeil is structured as a data-and-modelling pipeline rather than a single monolithic model, reflecting the fact that documenting a real-world crisis, forecasting on a proxy dataset, and measuring a privacy trade-off are three related but distinct problems. Public Libyan statistics and the Tetouan proxy dataset both feed a shared feature-engineering and chronological train/validation/test splitting stage; the resulting sequences are windowed into fixed-length lookbacks and passed to the Hybrid CNN-LSTM-Attention model for forecasting and anomaly detection, while a parallel branch applies differential-privacy noise to a synthetic Libyan-pattern series to measure the accuracy cost of formal privacy protection. All outputs converge in a single interactive dashboard.

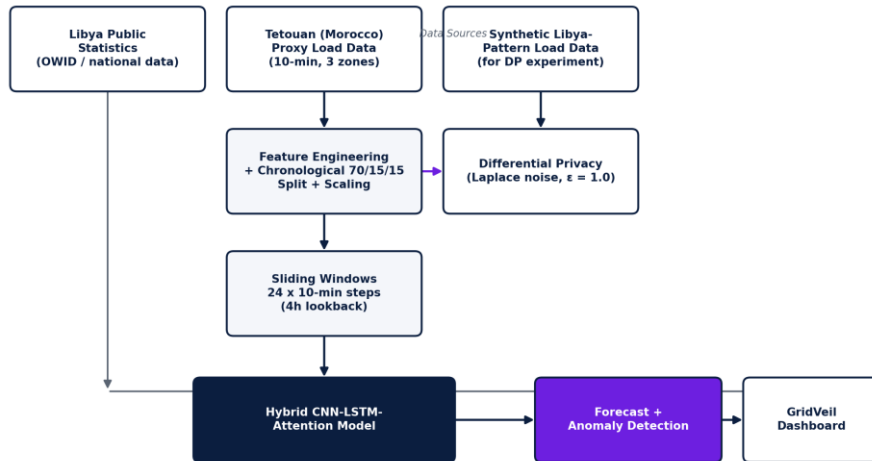


Figure 1.1 – GridVeil end-to-end data and modelling pipeline.

Tools and Technologies: Python (pandas, NumPy, scikit-learn) for data preprocessing and evaluation; TensorFlow/Keras for the Hybrid CNN-LSTM-Attention model; Google Colab with a T4 GPU for model training and the differential-privacy experiment; and a standalone HTML/CSS/JavaScript dashboard for presenting all results without requiring a server or database.

1.6.4 Implementation Details

The forecasting model was built as a single Keras functional-API network (see Section 5.2 for the full architecture) and trained with the Adam optimiser against a mean-squared-error loss, using early stopping (patience of 10 epochs on validation loss, restoring the best weights) and a model checkpoint that retains only the best-validation-loss version of the network. Training used a batch size of 64 for up to 50 epochs; in practice, early stopping halted training after 12 epochs once validation loss stopped improving. Feature and target scaling used a StandardScaler fitted only on the training split, then applied unchanged to the validation and test splits, to avoid any information leakage from future data into the training process. The anomaly-detection threshold was computed as the 95th percentile of the model's absolute error on the training set, rather than a fixed percentage of the load value, so that the threshold adapts automatically to each zone's actual error behaviour.

1.6.5 Evaluation Methodology

The forecasting model was evaluated using four standard regression metrics on the held-out chronological test split – Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE) and the coefficient of determination (R^2) – and compared directly against a Random Forest baseline trained and evaluated on the same 70/15/15 chronological split. Anomaly detection was evaluated by counting how many test-set observations exceeded the training-derived error threshold. The differential-privacy experiment was evaluated by comparing the MAPE of a model trained on the original synthetic Libyan-pattern data against the MAPE of an equivalent model trained on the same data after Laplace noise ($\epsilon = 1.0$) was applied, isolating the accuracy cost attributable to the privacy mechanism alone.

CHAPTER II – INITIATION

2.1 Feasibility Study

The project relies entirely on public datasets and pre-existing, well-established modelling techniques (CNN, LSTM, attention mechanisms, Random Forest and Laplace-mechanism differential privacy), orchestrated through a standard Python data-science stack and trained on a freely available Google Colab T4 GPU. Because no proprietary or licensed data, paid API, or specialised hardware is required, the technical feasibility of the project is high, and the main constraint is the availability and documentation quality of suitable public data rather than computational cost.

- Technical feasibility: all libraries used (pandas, NumPy, scikit-learn, TensorFlow/Keras, Matplotlib) are open-source and well-documented; training a compact hybrid model on ~52,000 rows of 10-minute data is comfortably within the memory and compute budget of a free-tier Colab GPU session.
- Operational feasibility: the pipeline is modular (separate preprocessing, model-definition and training scripts), which allowed the team to iterate on feature engineering and model architecture independently and to hand off final artefacts (trained model, loss curve, dashboard) without requiring every team member to re-run training.
- Economic feasibility: the entire prototype was built at zero monetary cost using free-tier compute and public data, with the main investment being engineering time rather than infrastructure or licensing spend.

2.2 Requirements Gathering

Requirements were derived from three sources: (1) a review of publicly reported facts about Libya's electricity sector (installed capacity, utility structure, distribution losses, renewable targets) to determine what the documentation layer of the dashboard needed to communicate; (2) a review of established time-series load-forecasting literature and benchmark datasets to select a methodologically defensible proxy dataset and a reasonable baseline/challenger model pairing (Random Forest vs. Hybrid CNN-LSTM-Attention); and (3) explicit privacy and ethics requirements set by the team from the outset – no personally identifiable or individually attributable data at any stage, and an explicit, visible distinction in the final dashboard between “real Libya context” and “proxy forecasting data.” These translated into the functional and non-functional requirements listed in Chapter III.

CHAPTER III – ANALYSIS

3.1 Software Requirement Specification

GridVeil shall ingest and preprocess public electricity-related datasets (Libyan national statistics, Tetouan proxy load data, and a synthetic Libyan-pattern series); shall engineer calendar- and behaviour-based features from timestamped consumption data; shall train and evaluate a Hybrid CNN-LSTM-Attention forecasting model and a Random Forest baseline on a strict chronological split; shall compute a data-driven anomaly-detection threshold and flag anomalous observations; shall apply a differential-privacy mechanism to a synthetic dataset and measure its accuracy impact; and shall present all of the above through a self-contained, browser-based dashboard requiring no server-side component.

3.2 Functional Requirements

- The system shall load and validate the chronological integrity (monotonicity, fixed 10-minute interval, no missing values) of the Tetouan proxy dataset before use.
- The system shall engineer calendar and behavioural features (hour, day of week, season, weekend/business-hour/peak-hour flags, etc.) from each record's timestamp.
- The system shall split all data chronologically into training (70%), validation (15%) and test (15%) sets, without shuffling, to prevent temporal leakage.
- The system shall scale features and targets using statistics computed only from the training split.
- The system shall construct sliding input windows of 24 consecutive 10-minute steps (4 hours) to forecast the next-hour load for three consumption zones.
- The system shall train a Hybrid CNN-LSTM-Attention model and a Random Forest baseline on identical data splits and report MAE, RMSE, MAPE and R^2 for both.
- The system shall compute an anomaly threshold from the 95th percentile of training absolute error and flag test-set observations exceeding it.
- The system shall apply Laplace-mechanism differential privacy ($\epsilon = 1.0$) to a synthetic Libyan-pattern dataset and report the resulting change in forecasting MAPE.
- The system shall present Libya context statistics, forecasting results, anomalies and privacy results through an interactive HTML dashboard.

3.3 Nonfunctional Requirements

- Reproducibility: all preprocessing and training steps must be deterministic and reproducible given the same public data sources.
- Transparency: every use of a proxy or synthetic dataset must be explicitly labelled as such in both the documentation and the dashboard.
- Privacy: no personally identifiable or individually attributable consumption data may be used at any stage of the project.
- Portability: the dashboard must run as a static HTML file in any modern browser, with no server, database or internet connection required at view time.
- Auditability: model artefacts (trained weights, loss curve) must be retained so that reported metrics can be independently verified.

3.4 Introduction to Analysis

The analysis in this chapter treats GridVeil as two connected sub-systems sharing a common data layer: a forecasting/anomaly-detection sub-system built on the Tetouan proxy dataset, and a privacy sub-system built on the synthetic Libyan-pattern dataset. Both are analysed here in terms of their actors, use cases and processing flow, before Chapter IV details their internal design.

3.5 System Scenario

A representative end-to-end scenario: a Content/Data Analyst prepares the Tetouan proxy dataset → the preprocessing stage validates timestamps and engineers features → the data is split chronologically and scaled → sliding windows are built → the Hybrid CNN-LSTM-Attention model and the Random Forest baseline are trained and evaluated on identical splits → the anomaly threshold is computed from training error and applied to the test set → in parallel, the synthetic Libyan-pattern dataset is perturbed with Laplace noise and a comparable model is trained to quantify the privacy-utility trade-off → all results (Libya context statistics, model comparison, anomalies, privacy trade-off) are rendered in the GridVeil dashboard for a Reviewer to inspect.

3.6 Use Case Diagram

Primary actors: Data/ML Engineer (prepares data, trains and evaluates models), Reviewer/Stakeholder (views the dashboard and interprets results). Main use cases: Prepare and Validate Dataset, Engineer Features, Train Forecasting Model, Evaluate Model Against Baseline, Detect Anomalies, Run Privacy Experiment, View Dashboard. The Reviewer interacts only with the “View Dashboard” use case, which in turn depends on all upstream data-preparation and modelling use cases having already produced their output artefacts (trained model, loss curve, computed metrics).

3.7 Activity Diagram

The activity flow mirrors the pipeline in Figure 1.1: (1) load and validate raw data → (2) engineer calendar/behavioural features → (3) chronologically split into train/validation/test → (4) fit scalers on the training split only and transform all splits → (5) build 24-step sliding windows → (6) train the Hybrid CNN-LSTM-Attention model with early stopping and checkpointing → (7) evaluate on the test split and compute MAE/RMSE/MAPE/ R^2 → (8) compute the anomaly threshold from training error and flag test anomalies → (9) in parallel, apply differential privacy to the synthetic dataset and re-train/evaluate → (10) publish all metrics and visualisations into the dashboard.

CHAPTER IV – DESIGN

4.1 Data Model

GridVeil does not use a relational database; instead, each dataset is a flat, versioned CSV file acting as the system's data model. The core forecasting dataset (Tetouan) uses the following schema:

- Datetime – timestamp at 10-minute resolution (index field, strictly monotonic, no gaps).
- Temperature, Humidity, WindSpeed, GeneralDiffuseFlows, DiffuseFlows – exogenous weather/solar features.
- PowerConsumption_Zone1, PowerConsumption_Zone2, PowerConsumption_Zone3 – the three forecasting targets.
- Derived features (engineered at runtime) – hour, minute, dayofweek, dayofmonth, dayofyear, weekofyear, month, quarter, season, is_weekend, is_working_day, is_business_hours, is_peak_hour, is_month_start/end, is_quarter_start/end, minute_of_day, minute_of_week.

The Libya context dataset uses a normalised long format (Technology/Plant, Year, Value, Parameter), which allows capacity, cost and policy figures from different sources to be combined and filtered consistently for the dashboard's charts.

4.2 Sequence Diagram

Interaction order for a single training run: Scheduler/Notebook → `data_preprocessing.load_data()` (reads CSV, parses and sorts timestamps) → `verify_time_steps()` (checks monotonicity, fixed interval, missing values) → `create_features()` (adds calendar/behavioural columns) → chronological split → `StandardScaler.fit()` on train only → `create_sliding_windows()` for each split → `model.build_hybrid_model()` → `model.fit()` with `EarlyStopping/ModelCheckpoint` callbacks → `model.predict()` on the test split → metric functions (MAE, RMSE, MAPE, R^2) → anomaly threshold computed from training-set errors → results and loss-curve plot written to disk for the dashboard to consume.

4.3 Component Model

The system is organised into five loosely coupled components:

- `data_preprocessing.py` – loading, validation, feature engineering, chronological splitting, scaling and windowing.
- `model.py` – definition of the Hybrid CNN-LSTM-Attention architecture and its training callbacks.
- `train.py` – orchestrates preprocessing, training, evaluation, anomaly-threshold computation and loss-curve plotting.
- Differential-privacy module (Colab) – applies Laplace noise to the synthetic Libyan-pattern dataset and re-runs an equivalent training/evaluation cycle for comparison.
- `dashboard/gridveil.html` – a standalone front-end component that renders Libya context statistics, model-comparison results, anomaly flags and the privacy trade-off, reading only static, pre-computed results (no live backend).

4.4 Deployment Diagram

Model training and the differential-privacy experiment were executed in Google Colab using a T4 GPU runtime, chosen for its free availability and sufficient memory for the dataset size involved. The reusable output artefacts – the trained model file (`best_gridveil_model.keras`) and the training-loss plot – are

committed back into the repository under a results/ directory rather than requiring retraining on every machine that views the project. The dashboard is a static HTML/CSS/JavaScript file with no server dependency, so it can be opened directly in any modern browser on any machine, independent of where training took place.

CHAPTER V – DEVELOPMENT AND TESTING

5.1 Data Sources and Preparation

GridVeil does not use a traditional database; four flat public/derived datasets serve as its data layer, each with a distinct, explicitly documented role:

Dataset	Role	Key characteristics
Libya national statistics	Real-world context	Installed capacity by technology/year, plant-level capacity and lifetime, capital-cost trajectories 2015–2050, renewable-policy targets
Tetouan (Morocco) load data	Forecasting proxy	52,417 rows, 10-minute resolution from Jan 2017, 3 consumption zones + 5 weather/solar features
Synthetic Libya-pattern data	Privacy experiment	Hourly, documented seasonal/diurnal load shape; no real measurements
Engineered features	Model input	16 calendar/behavioural features derived from each timestamp

This table corresponds to Section 1.6.2; installed-capacity figures for Libya as of the most recent year in the public dataset (2018) are summarised below as an example of the real-world context data used in the dashboard.

Generation technology	Installed capacity (MW), 2018
Gas Power Plant (CCGT)	6,761.57
Gas Power Plant (SCGT)	2,120.00
Oil-Fired Gas Turbine (SCGT)	191.43
Off-grid Solar PV	5.11

5.2 System Development

The forecasting model was implemented as a Keras functional-API network named GridVeil_CNN_LSTM_Attention, combining local pattern extraction, sequence memory and attention-based weighting in a single trainable pipeline:

- A 1D convolution (64 filters, kernel size 3, ReLU activation) extracts short-term local patterns from adjacent 10-minute time steps.
- Max-pooling (pool size 2) reduces temporal resolution while retaining the strongest local signals.
- An LSTM layer (64 units, returning full sequences) encodes longer-term temporal dependencies across the pooled sequence.
- A self-attention layer re-weights the encoded time steps by relative importance, followed by global average pooling to aggregate them into a single vector.

- A dense hidden layer (32 units, ReLU) and a final linear output layer (3 units) produce the next-hour forecast for all three consumption zones simultaneously.

The model was compiled with the Adam optimiser and mean-squared-error loss, and trained with early stopping (patience 10, monitoring validation loss, restoring best weights) and a checkpoint callback that saves only the best-validation-loss version to `best_gridveil_model.keras`. Training used a maximum of 50 epochs at batch size 64; early stopping halted training after 12 epochs. As a challenger comparison, a Random Forest baseline was trained on the same engineered features and identical chronological split (“Step 7” in the project’s evaluation sequence).

5.3 Testing Plan

The system was evaluated through manual functional, integration and end-to-end testing, consistent with the applied, integration-focused nature of the project rather than a purely model-centric research study.

- Functional tests verified that each pipeline stage – data validation, feature engineering, chronological splitting, scaling, windowing, model training and anomaly-threshold computation – performed correctly in isolation.
- Integration tests verified correct, leakage-free data flow between preprocessing and modelling stages, including confirming that scalers were fitted only on training data and that windows did not cross split boundaries.
- End-to-end tests executed the full pipeline from raw CSV ingestion through to final metric computation and loss-curve generation, and separately validated the differential-privacy branch from synthetic data through to its own MAPE comparison.

5.4 Testing Results

The Hybrid CNN-LSTM-Attention model was evaluated on the held-out chronological test split and directly compared against the Random Forest baseline trained on the same data and split:

Metric	Random Forest baseline	Hybrid CNN-LSTM-Attention
Lookback window	4 hours (24 samples)	4 hours (24 samples)
Train/validation/test split	70/15/15 (chronological)	70/15/15 (chronological)
Test MAPE	5.77%	2.69%

The anomaly-detection layer, using a threshold derived from the 95th percentile of training-set absolute error, produced the following result on the test set:

Metric	Value
Dynamic anomaly threshold (95th percentile of training error)	0.421
Anomalies detected in test set	7,834

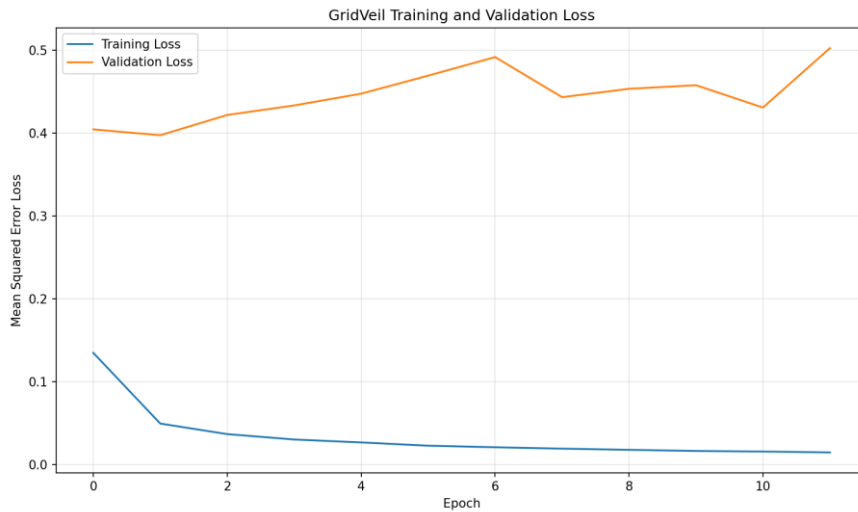


Figure 5.1 – Training and validation loss curve, Hybrid CNN-LSTM-Attention model (early stopping triggered after 12 epochs).

The differential-privacy experiment measured the accuracy cost of protecting the synthetic Libyan-pattern dataset with Laplace-mechanism noise at a privacy budget of $\epsilon = 1.0$, applied before training:

Configuration	Test MAPE
Original synthetic data (no privacy noise)	5.57%
Privacy-protected data (Laplace noise, $\epsilon = 1.0$)	7.24%

At $\epsilon = 1.0$, protecting the data increased forecasting error by roughly 30% in relative terms (5.57% $\rightarrow$ 7.24% MAPE). Because the Laplace noise scale is fixed relative to the series' mean regardless of ϵ , this single-point measurement is a useful first indication of the privacy-utility trade-off rather than a complete curve; mapping accuracy loss across a wider range of ϵ values (for example 0.1 to 5) would show more precisely where the cost becomes acceptable for a given operational use case.

CHAPTER VI – CONCLUSION

6.1 Summary

This project addressed the challenge of analysing and forecasting Libya's electricity crisis in the complete absence of public, granular national consumption data. GridVeil combines real public statistics documenting Libya's generation, distribution and reliability problems with a Hybrid CNN-LSTM-Attention forecasting and anomaly-detection pipeline built and validated on a public, climatically comparable proxy dataset from Tetouan, Morocco, and a differential-privacy experiment quantifying the accuracy cost of formally protecting consumption data. All results are integrated into a single, self-contained dashboard.

6.2 Conclusion

The Hybrid CNN-LSTM-Attention model achieved a test MAPE of 2.69%, substantially outperforming a Random Forest baseline at 5.77% MAPE on an identical chronological split, demonstrating that combining convolutional, recurrent and attention mechanisms yields tangible accuracy gains for short-horizon electricity load forecasting. The data-driven anomaly-detection layer, using an adaptive threshold derived from the model's own training error, flagged 7,834 anomalous observations in the test set without relying on arbitrary fixed cutoffs. The differential-privacy experiment showed that a meaningful, quantifiable privacy guarantee ($\epsilon = 1.0$) comes at a real but bounded accuracy cost – a roughly 30% relative increase in forecasting error – giving project stakeholders a concrete, evidence-based starting point for future privacy-budget decisions. Taken together, these results demonstrate that a scientifically credible, ethically grounded and privacy-safe forecasting methodology can be built and validated today, entirely from public data, and stands ready for direct transfer to Libya's own grid once equivalent local data becomes publicly available.

6.3 Suggestions

- Map the full differential-privacy trade-off curve across a range of ϵ values (e.g., 0.1 to 5) to identify the operating point that balances accuracy and privacy for a specific use case.
- Extend the anomaly-detection layer with automated regression testing and a labelled evaluation set, to move from unusual-behaviour flagging toward validated outage or fault detection.
- Apply the validated methodology directly to Libya's own grid data as soon as GECOL or another authorised body makes equivalent granular data available, replacing the Tetouan proxy without changing the underlying pipeline.
- Extend the forecasting horizon beyond the current next-hour prediction to support day-ahead generation planning.
- Reduce reliance on a single proxy dataset by validating the methodology against additional comparable regional datasets before any real-world deployment.
- Add live analytics on dashboard usage and model drift monitoring if the system is ever connected to a live or periodically updated data feed.